

CS621/CSL622

Quantum Computing For Computer Scientists

The Leap from Classical to Quantum

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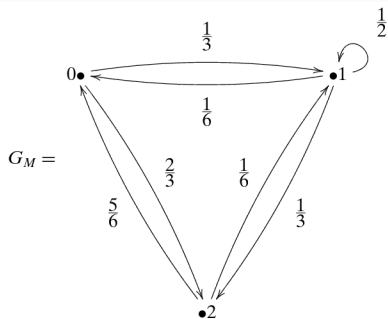
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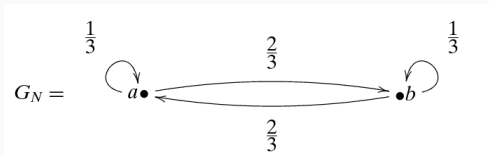
Assembling Systems

Dealing with Composite Systems

Assembling Classical Probabilistic Systems



$$M = \begin{bmatrix} 0 & \frac{1}{6} & \frac{5}{6} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}.$$



$$N = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Two-marble System: G_M for red-marble, G_N for blue marble

How does a state for a two-marble system look?

- There are $3 \times 2 = 6$ possible states of the combined system.
Why?
 - The red marble can be on one of three vertices and
 - The blue marble can be on one of two vertices,
- This is the **tensor product**¹ of a 3-by-1 vector with a 2-by-1 vector

Example (A typical combined state)

$$X = \begin{matrix} 0a \\ 0b \\ 1a \\ 1b \\ 2a \\ 2b \end{matrix} \begin{bmatrix} \frac{1}{18} \\ 0 \\ \frac{2}{18} \\ \frac{1}{3} \\ 0 \\ \frac{1}{2} \end{bmatrix}$$

← How to interpret this?

¹More on tensor product in up-coming lectures.

Dynamics of a Composite System

- For a system to go from state ij to a state $i'j'$ we must multiply the probability of going from state i to state i' with the probability of going from state j to state j' .

$$ij \xrightarrow{M[i',i] \times N[j',j]} i'j' \text{ [Provided systems are independent]}$$

- What is the probability of going from state $1a$ to state $2b$?

$$X = \begin{matrix} 0a \\ 0b \\ 1a \\ 1b \\ 2a \\ 2b \end{matrix} \begin{bmatrix} \frac{1}{18} \\ 0 \\ \frac{2}{18} \\ \frac{1}{3} \\ 0 \\ \frac{1}{2} \end{bmatrix}$$

$$M = \begin{bmatrix} 0 & \frac{1}{6} & \frac{5}{6} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}.$$

$$N = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

- What does it mean for the entire system? **Tensor Product**

Tensor Product $\otimes$ Captures Dynamics

$$M \otimes N = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 0 \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & \frac{1}{6} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & \frac{5}{6} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} \\ \frac{1}{3} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & \frac{1}{2} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & \frac{1}{6} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} \\ \frac{2}{3} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & \frac{1}{3} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} & 0 \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} \end{bmatrix}$$

$$= \begin{matrix} & \begin{matrix} 0a & 0b & 1a & 1b & 2a & 2b \end{matrix} \\ \begin{matrix} 0a \\ 0b \\ 1a \\ 1b \\ 2a \\ 2b \end{matrix} & \begin{bmatrix} 0 & 0 & \frac{1}{18} & \frac{2}{18} & \frac{5}{18} & \frac{10}{18} \\ 0 & 0 & \frac{2}{18} & \frac{1}{18} & \frac{10}{18} & \frac{5}{18} \\ \frac{1}{9} & \frac{2}{9} & \frac{1}{6} & \frac{2}{6} & \frac{1}{18} & \frac{2}{18} \\ \frac{2}{9} & \frac{1}{9} & \frac{2}{6} & \frac{1}{6} & \frac{2}{18} & \frac{1}{18} \\ \frac{2}{9} & \frac{4}{9} & \frac{1}{9} & \frac{2}{9} & 0 & 0 \\ \frac{4}{9} & \frac{2}{9} & \frac{2}{9} & \frac{1}{9} & 0 & 0 \end{bmatrix} \end{matrix}$$

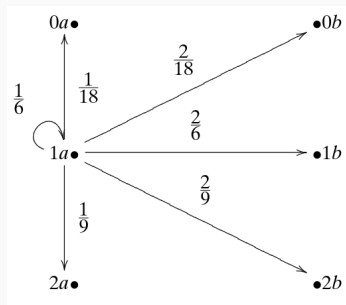
- The graph that corresponds to this matrix, $G_M \times G_N$ - called the **Cartesian product** of two weighted graphs has 28 weighted arrows.

- Find the matrix and the graph that correspond to $N \otimes N$.

$$N = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Visualizing $M \otimes N$ (Third Column)

	$0a$	$0b$	$1a$	$1b$	$2a$	$2b$
$0a$	0	0	$\frac{1}{18}$	$\frac{2}{18}$	$\frac{5}{18}$	$\frac{10}{18}$
$0b$	0	0	$\frac{2}{18}$	$\frac{1}{18}$	$\frac{10}{18}$	$\frac{5}{18}$
$1a$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{1}{18}$	$\frac{2}{18}$
$1b$	$\frac{2}{9}$	$\frac{1}{9}$	$\frac{2}{6}$	$\frac{1}{6}$	$\frac{2}{18}$	$\frac{1}{18}$
$2a$	$\frac{2}{9}$	$\frac{4}{9}$	$\frac{1}{9}$	$\frac{2}{9}$	0	0
$2b$	$\frac{4}{9}$	$\frac{2}{9}$	$\frac{2}{9}$	$\frac{1}{9}$	0	0



Attempt this again

- What is the probability of going from state $1a$ to state $2b$?

Composite System State Evolution

- Tensor product of the matrices will then act on the tensor product of the vectors.

$$\begin{array}{c} \mathbf{0a} \\ \mathbf{0b} \\ \mathbf{1a} \\ \mathbf{1b} \\ \mathbf{2a} \\ \mathbf{2b} \end{array} \begin{bmatrix} \mathbf{0a} & \mathbf{0b} & \mathbf{1a} & \mathbf{1b} & \mathbf{2a} & \mathbf{2b} \\ 0 & 0 & \frac{1}{18} & \frac{2}{18} & \frac{5}{18} & \frac{10}{18} \\ 0 & 0 & \frac{2}{18} & \frac{1}{18} & \frac{10}{18} & \frac{5}{18} \\ \frac{1}{9} & \frac{2}{9} & \frac{1}{6} & \frac{2}{6} & \frac{1}{18} & \frac{2}{18} \\ \frac{2}{9} & \frac{1}{9} & \frac{2}{6} & \frac{1}{6} & \frac{2}{18} & \frac{1}{18} \\ \frac{2}{9} & \frac{4}{9} & \frac{1}{9} & \frac{2}{9} & 0 & 0 \\ \frac{4}{9} & \frac{2}{9} & \frac{2}{9} & \frac{1}{9} & 0 & 0 \end{bmatrix} \begin{array}{c} \mathbf{0a} \\ \mathbf{0b} \\ \mathbf{1a} \\ \mathbf{1b} \\ \mathbf{2a} \\ \mathbf{2b} \end{array} \begin{bmatrix} \frac{1}{18} \\ 0 \\ \frac{2}{18} \\ \frac{1}{3} \\ 0 \\ \frac{1}{2} \end{bmatrix}$$

What happens to Composite Systems in Quantum Theory?

- Similar to probabilistic systems but has more!
 - The states of two separate systems can be combined using the tensor product of two vectors
 - The changes of two systems are combined by using the tensor product of two matrices
 - The tensor product of the matrices will then act on the tensor product of the vectors

Whats More?

Entangled States

- In the quantum world there are many more possible states than just states that can be **combined from smaller ones**.
 - There can be states that cannot be expressed as the tensor product of the smaller states
 - These are the more interesting ones → **Entangled States**
- Similarly, arguments are there for operations on a combined quantum system

- How many vertices for one-bit with a marble?
- How many vertices for m-bits?
- Size of transition matrix?
- Can you see the **exponential growth**² in resources required?
- Can a classical computer simulate such a system?

The Quantum Computer

A prospective quantum computer, with its inherent ability to perform massive parallel processing, might be able to accomplish the task.

²This exponential growth is actually one of the main reasons Richard Feynman started talking (Feynman, 1982) about quantum computing in the first place.

- A composite system is represented by the Cartesian product of the transition graphs of its subsystems.
- If two matrices act on the subsystems independently, then their tensor product acts on the states of their combined system.
- There is an exponential growth in the amount of resources needed to describe larger and larger composite systems